Decision Tree
&
Random Forest

Iris Flower Dataset Classification

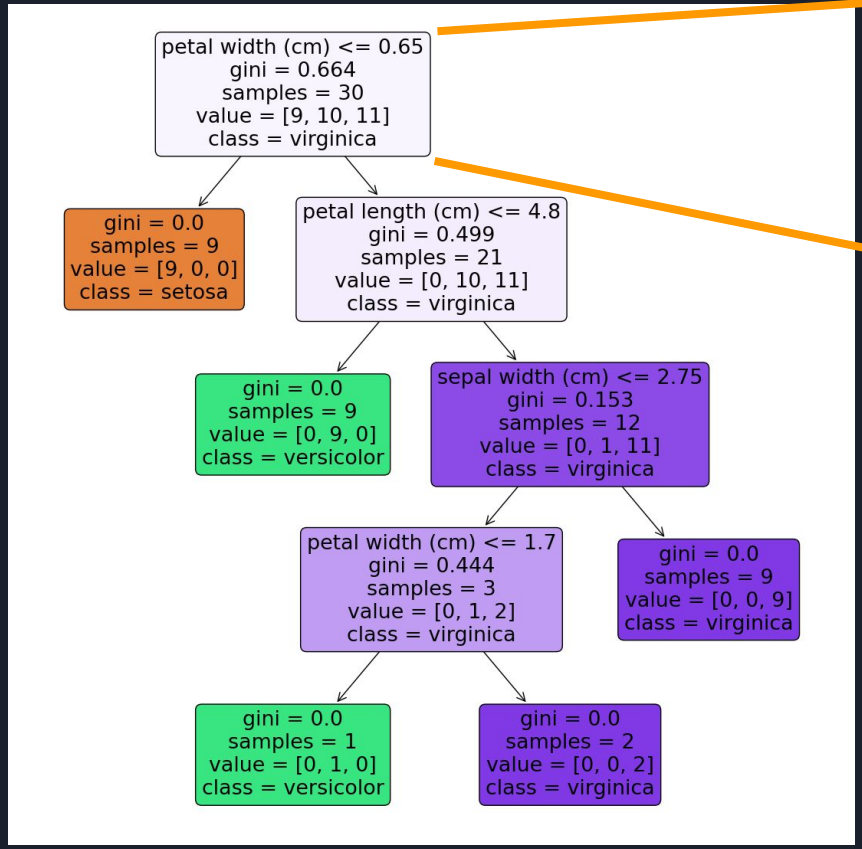
Overview

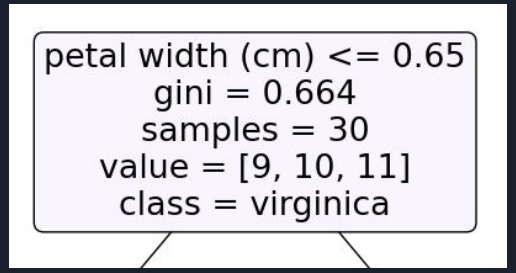
- Decision Tree is used in supervised learning for both classification and regression tasks
  - It builds a tree-like structure to make decisions or predictions based on input data
  - They are intuitive and interpretable, making them valuable tools for a variety of applications
- Random Forest is an ensemble learning method in machine learning
  - Meaning , it uses multiple decision trees in training and prediction
  - Helps address some of the limitations of single decision trees, such as overfitting.

Classify with Decision Tree & Random Forest

- Iris Flower dataset will be used for the classifications
- A selection of 4 parameters in each of Decision Tree and Random Forest algorithms will be tested against the default ones
- Split: Train=20%(30 samples), Test=80%(120 samples)
  - Due to DT and RF are really good at classifying Iris dataset, the reason for the ratio is to make parameter changes more obvious
  - The same split dataset will be used for all training tasks

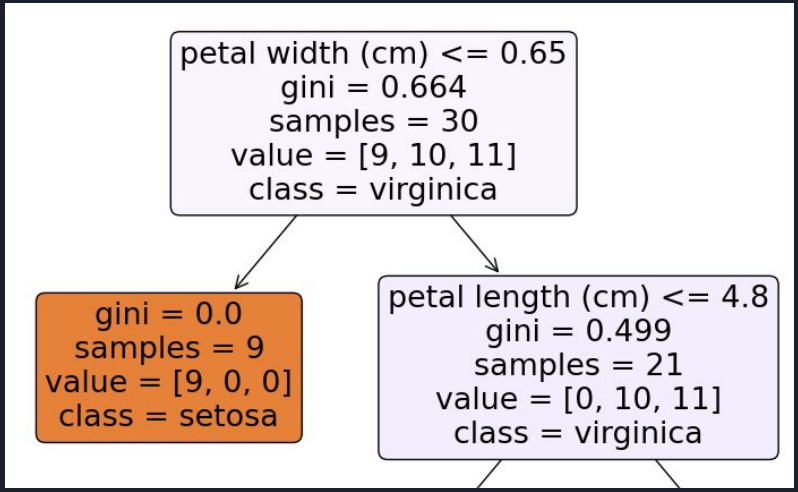
Decision Tree Plot Overview





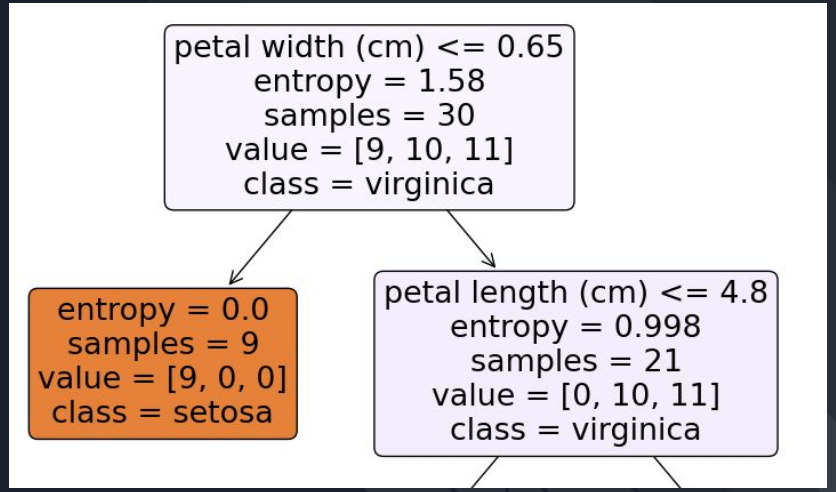
Color -> Class
Depth = 4

criterion=gini
(score: 0.94)



$$1 - \sum_{i=1}^{J} p_i^2$$

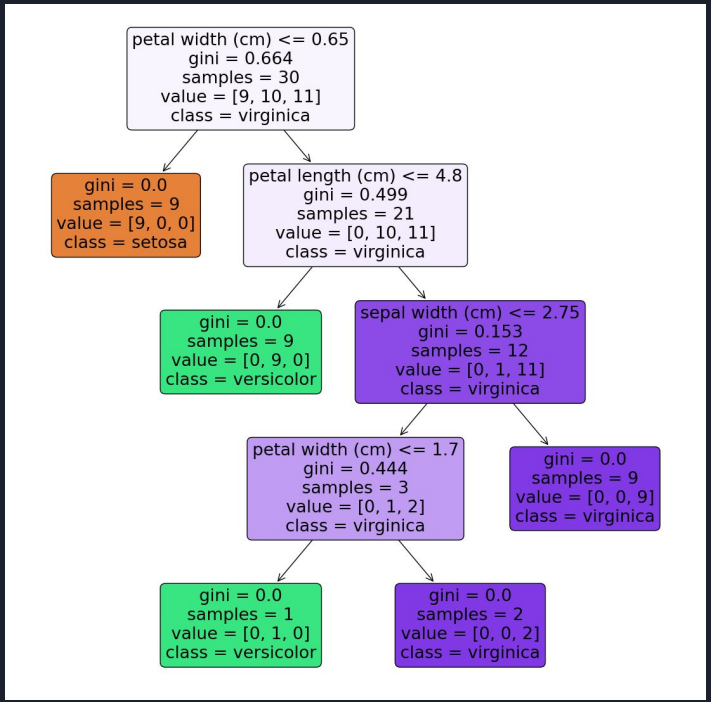
criterion=entropy
(score: 0.94)



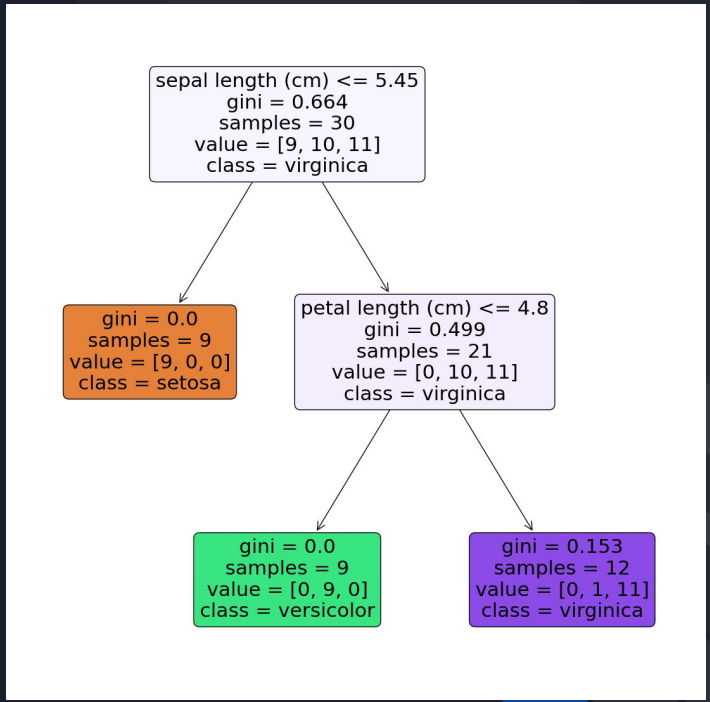
$$-\sum_{i=1}^{J} p_i \log_2 p_i$$

The function to measure the quality (impurity) of a split

max_depth=None
(score: 0.94)

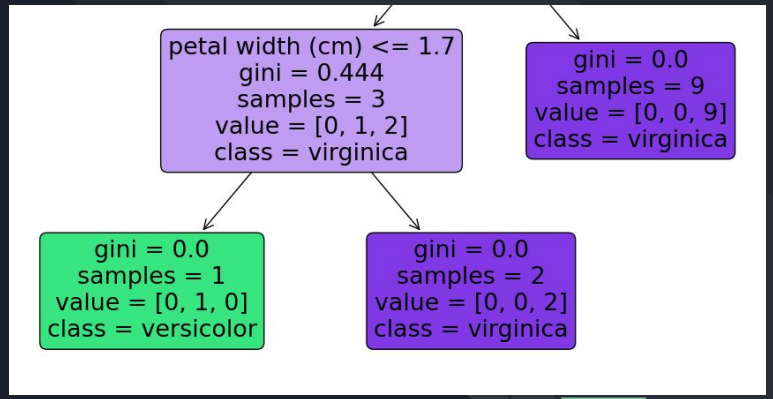


max_depth=2
(score: 0.86)

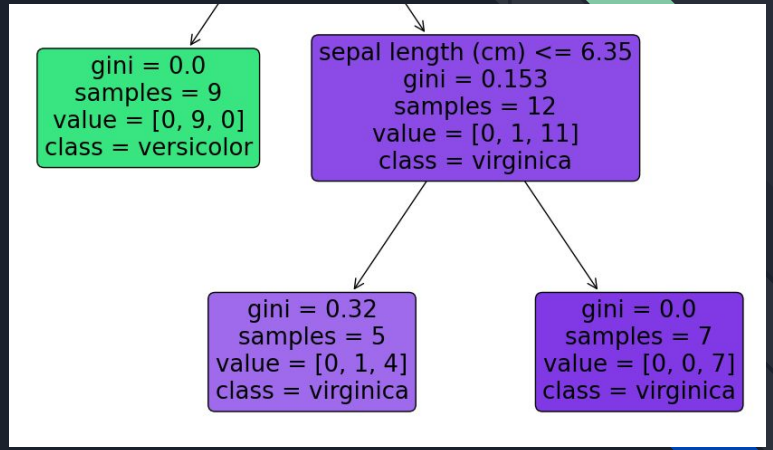


The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples

min_samples_leaf=1
(score: 0.94)



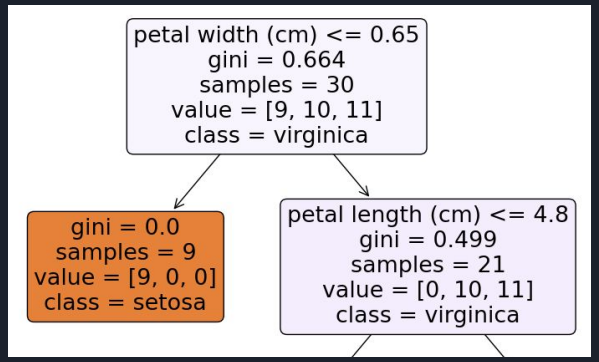
min_samples_leaf=5
(score: 0.86)

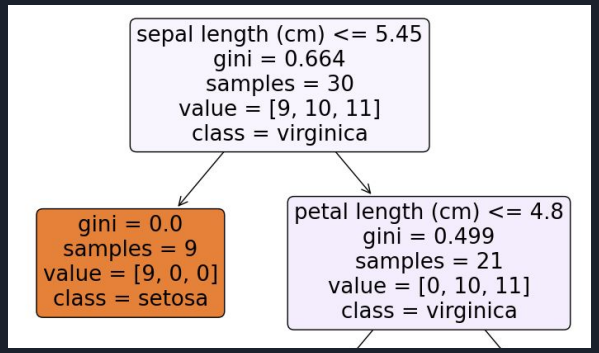


The minimum number of samples required to be at a leaf node

random_state=None

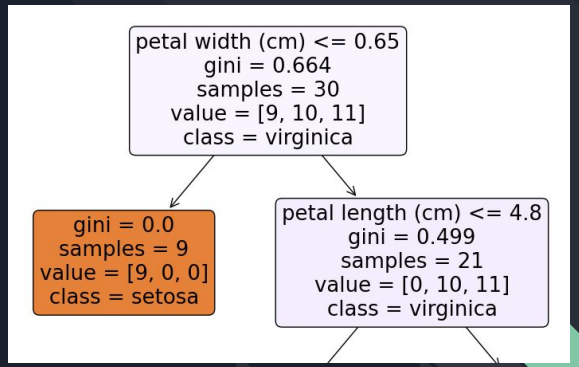
(score: 0.94, 0.85)

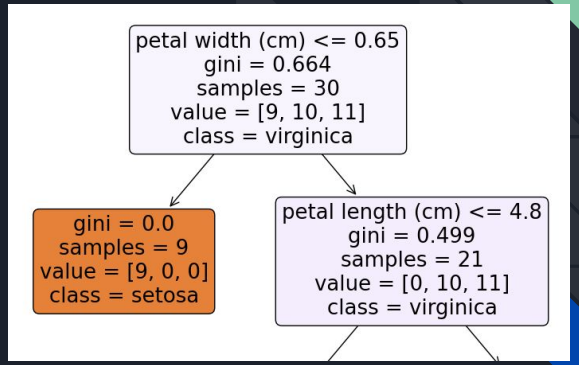




random_state=0

(score: 0.93, 0.93)

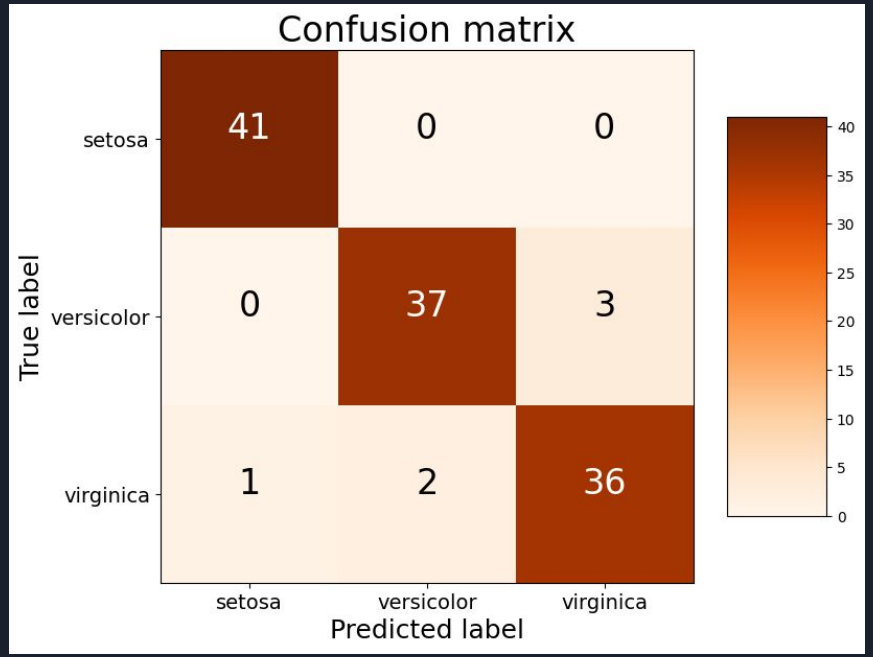




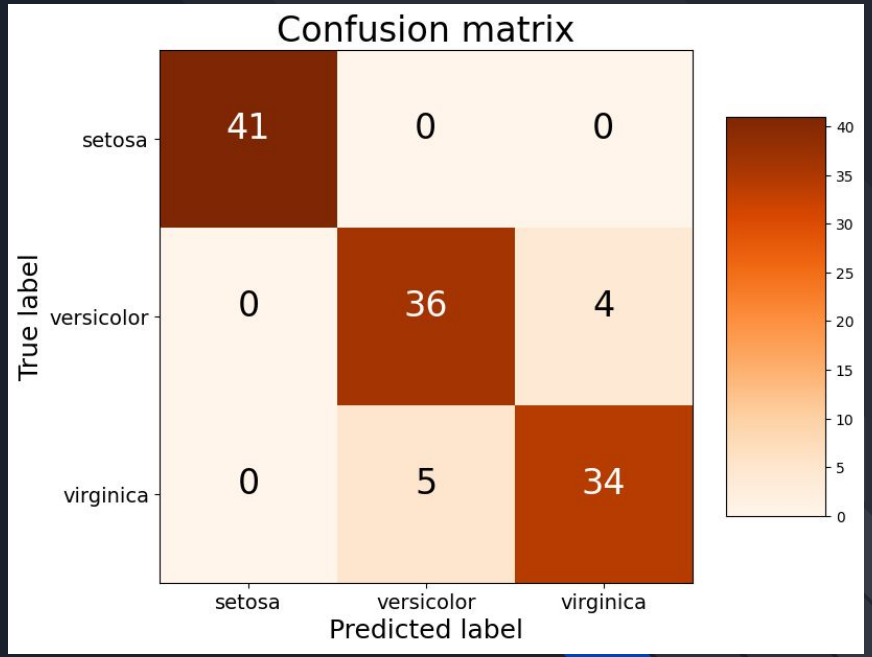
Controls the randomness of the estimator

Random Forest

n_estimators=100

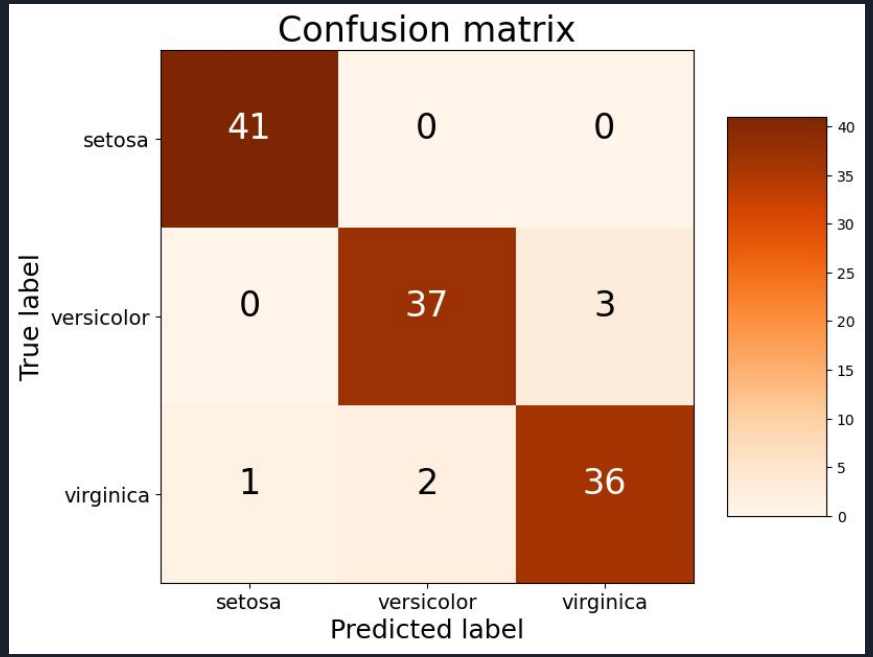


n_estimators=1

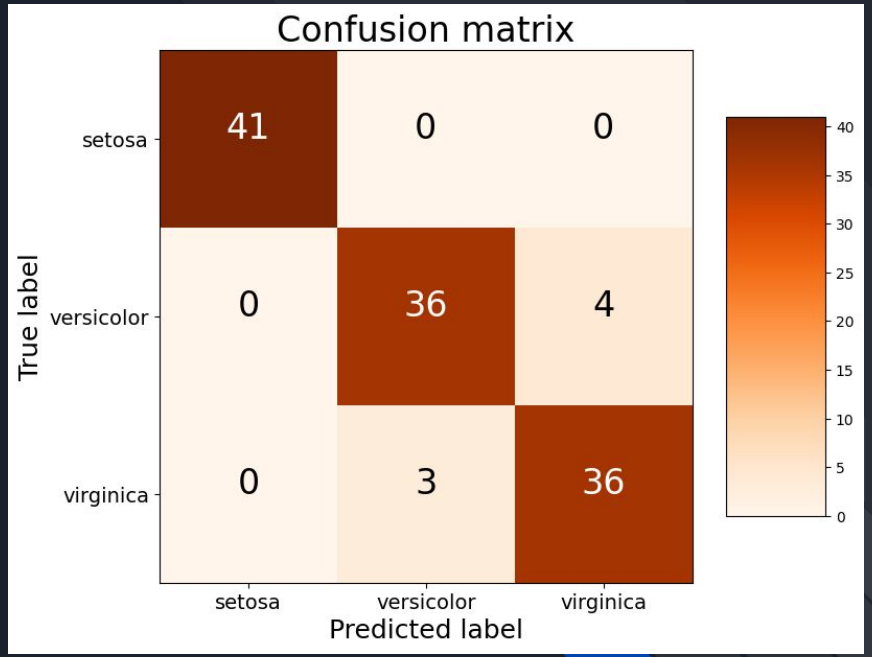


The number of (decision) trees in the forest

max_features=sqrt

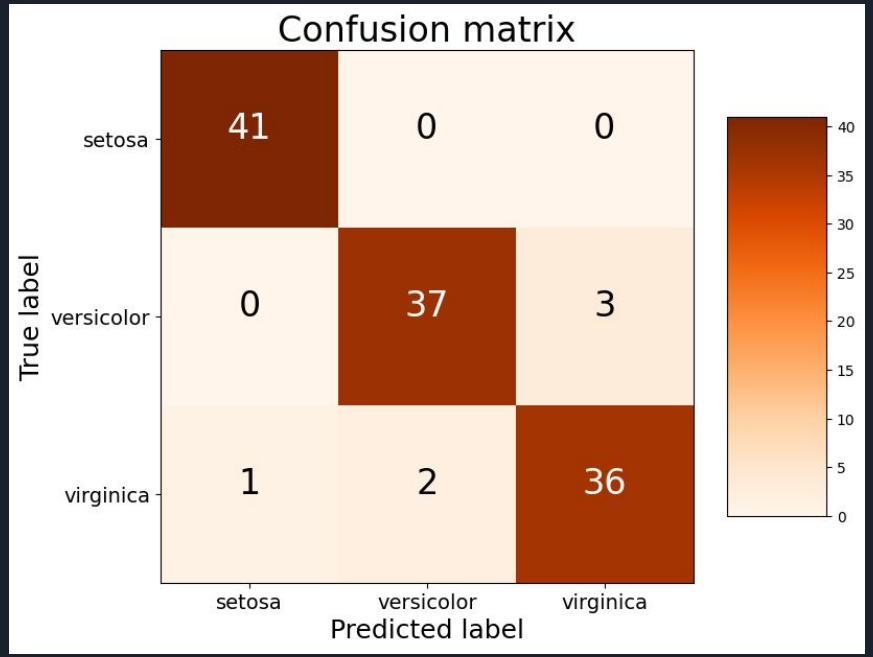


max_features=None
(max_features=n_features)

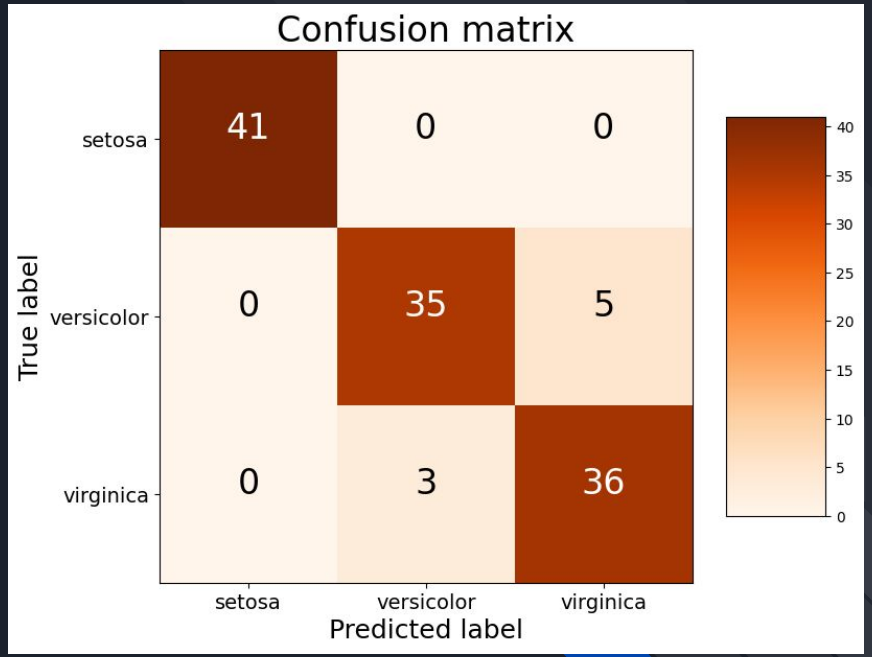


The number of features to consider when looking for the best split

bootstrap=True

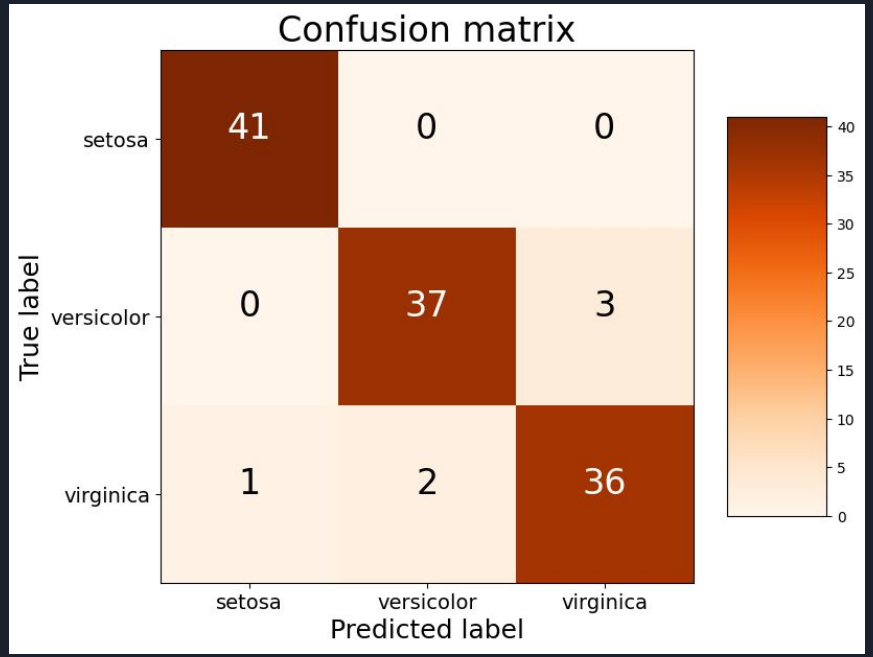


bootstrap=False

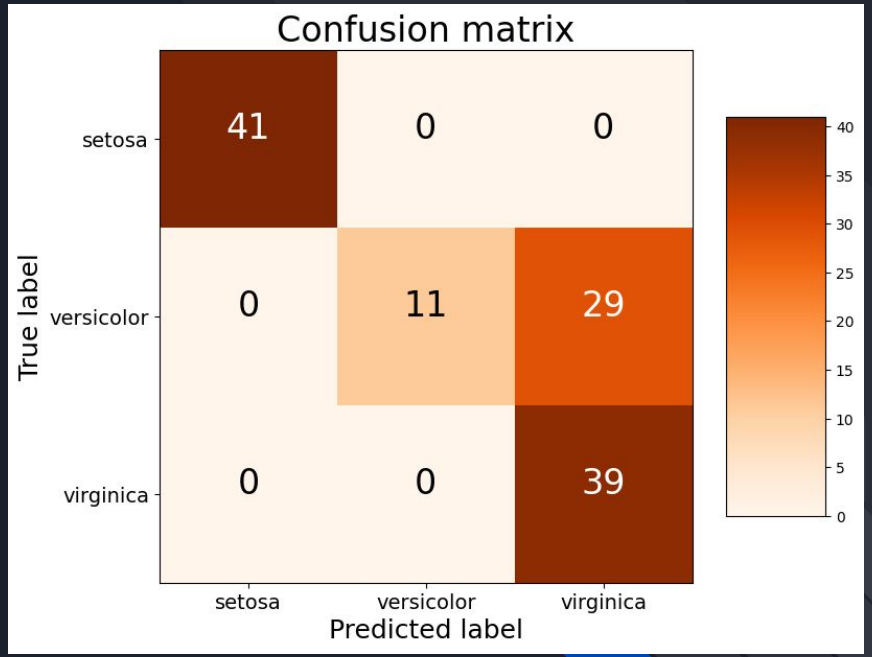


Whether bootstrap samples are used when building trees. If False, the whole dataset is used to build each tree

max_samples=None
(max_samples=X.shape[0])



max_samples=5



If bootstrap is True, the number of samples to draw from input to train each base estimator

End of Presentation